THz Access Points Deployment via Volumetric and Directional Optimization under Dynamic Blockages

Arpit Kumar, Arpit Bhomia, Ashish Ranjan, Abhigyan Sharma, and Anand M. Baswade
Department of Computer Science and Engineering, Indian Institute of Technology Bhilai, India
Email: {arpitk, arpitba, ashishr, abhigyans, anand}@iitbhilai.ac.in

***Abstract*—The Terahertz (THz) band used in 6G networks provides very high data speeds, but it is directional, highly sensitive to various types of loss, and vulnerable to dynamic human blockage. Traditional Methods for indoor deployment strategies for Access Point (APs) are normally based on a 2D plane, but this approach could be improved as they force the signals to travel through the crowd and the obstacles without considering whether the height of the AP placement really needs to consider that part of the crowd or not. In this paper, we propose a comprehensive, cost-effective 3D deployment strategy that aims to avoid these blockages. To avoid relying on suboptimal greedy strategies, we formulate it as a Mixed-Integer Linear Programming (MILP) model to guarantee optimal layouts. Our framework optimizes the vertical mounting elevation (Z-axis) and the directional azimuth angle of each AP, balancing line-of-sight path-loss trade-offs to route the beams above physical obstacles. Furthermore, we incorporate an empirical human crowd-density model, enabling the system to measure AP deployments based on stochastic crowd footfall. Simulations in a realistic indoor environment show that this joint optimization guarantees strict signal quality in congested areas, achieving up to a 48% reduction in required Access points compared to state-of-the-art heuristics.**

***Index Terms*—THz, Access Point Placement, Human Blockage, MILP, Optimization, 3D Propagation**

I. Introduction

As we move toward 6G networks, the demand for super-fast, multi-Terabit-per-second (Tbps) data rates has driven research into the Terahertz (THz) band. The THz band has a lot of free spectrum compared to traditional sub-6 GHz or millimeter-wave networks. This creates exceptional data capacities for indoor environments, such as smart offices and academic campuses.

However, using the THz band poses many challenges, as THz signals degrade rapidly in the atmosphere and are readily absorbed by atmospheric moisture. To compensate for this signal decay, transmitters must use highly focused, narrow beams. Since these beams require a clear, uninterrupted Line-of-Sight (LoS) to the user, they are vulnerable to physical blockages. In indoor settings, the most disruptive obstacle is moving people. Human bodies cause significant signal drops (often exceeding 20 dB) that lead to network outages.

Existing literature relies on 2D models in which each Access Point is fixed at the ceiling elevation and fails to jointly optimize directional azimuth orientations. Also, current models overlook the dynamics of human blockage, which causes severe signal drops.

This planar approach is inefficient because it ignores vertical Z-axis adjustments, optimal azimuth targeting, and dynamic blockage (e.g., humans or signal drops). As a result, it deploys a high number of Access Points. This paper uses the term "Access Points/Base Stations" to refer to the transmitting nodes. This term refers to the physical infrastructure that provides network coverage, bridging the gap between traditional Wi-Fi access points and indoor cellular base stations.

The main **contributions** of this paper are summarized as:

- We propose a 3D volumetric path loss model that accounts for THz free-space loss, molecular absorption, static losses, and dynamic human blockage.
- We define a deployment metric based on an empirical density Threshold (τ), allowing network operators to deploy APs based on densely populated areas.
- We adopt a highly directive antenna model that jointly optimizes the AP's azimuth orientation to enable the system to balance the trade-off between transmission range and spatial coverage without violating radiated energy conservation.
- We formulate the deployment problem as a Mixed-Integer Linear Programming (MILP) model to explicitly optimize the exact 3D spatial coordinates and azimuth orientations of THz APs, minimizing total number of APs deployed.

II. Related Work

The deployment of Terahertz (THz) communication networks depends primarily on addressing key propagation challenges, which has led to substantial research on channel modeling and deployment methodologies.

A. THz Channel Characterization

The fundamental physical characteristics of the THz channel have been characterized by Akyildiz *et al.* [5]. Their work provides an in-depth analysis of the THz band spanning 0.1–10 THz, identifying it as a key frontier for next-generation wireless communications capable of supporting Terabit-per-second (Tbps) links. The THz path loss is decomposed into two principal components: spreading loss, which accounts for signal attenuation due to the spatial expansion of the electromagnetic wave, and molecular absorption loss, caused by the interaction of THz radiation with atmospheric constituents such as water vapor. For distances beyond a few meters, the total path loss can exceed 100 dB . To compensate for this extreme attenuation, Akyildiz *et al.* emphasize the necessity of highly directional and high-gain antennas. Our work directly builds on these channel insights by incorporating both the

free space path loss and a frequency dependent molecular absorption coefficient $k(f)$ into a unified path loss model (Eq. 3), and by optimizing a directional flat top antenna model using the half-power beamwidth (Eq. 10).

B. Dynamic Blockage Modeling

Because THz signal lack the diffraction capabilities of sub-6 GHz signals, they are highly susceptible to dynamic blockages. Research into THz channel reliability has shown that human bodies act as full blockers [6], causing signal degradation often exceeding 20 dB. Unlike static obstacles whose positions can be determined in advance, human mobility introduces a stochastic component that state of the art deployment models typically ignore. Therefore, dynamic obstruction analysis that accounts for the presence of human blockage is an essential prerequisite for effective network planning. Our framework addresses this gap by introducing a normalized density metric derived from pedestrian datasets (Eq. 17).

C. AP Placement Methodologies

To address these propagation challenges, recent research has adopted various methods of AP positioning. Alameh *et al.* [7] investigated the optimal Line-of-Sight (LoS) coverage problem for multi AP millimeter-wave wireless LANs in indoor environments. For scenarios with random obstacle and random client locations (RORC), they reformulated the AP deployment problem as a *thinnest covering problem*, analytically deriving optimal ceiling-mounted AP positions for rectangular rooms by finding the placement that minimizes the maximum horizontal distance between any client and its nearest AP. For fixed-obstacle scenarios (FORC), they proposed a Shadowing-Elimination Search (SES) algorithm that sequentially places APs by iteratively minimizing the remaining shadowing regions caused by known obstacles. Their approach demonstrates that ceiling mounted APs maximize LoS probability, since the LoS probability increases monotonically with AP height, and their SES algorithm achieves near optimal LoS coverage with significant computational efficiency over brute-force methods.

Despite the strengths of this approach, several limitations remain. First, it operates exclusively in a 2D plane with APs fixed at a uniform ceiling height, and does not optimize over the vertical Z-axis dimension. Second, the path loss and blockage models are designed for 60 GHz millimeter-wave frequencies, which do not account for THz-specific phenomena such as molecular absorption loss. Third, their framework considers only static blockages from furniture and obstacles, without incorporating the stochastic effects of dynamic human mobility. Fourth, the antenna model does not jointly optimize the directional azimuth orientation of each AP. While heuristic and geometry based approaches such as these offer computational efficiency, they may not guarantee an optimal solution which is critical for number of APs deployed and maintaing a certain SINR threshold.

D. Directional Beam Steering and User Orientation

The role of beam directionality and user orientation in mmW/THz network coverage has been studied by Soorki *et al.* [8]. They proposed a framework for jointly optimizing AP deployment and beam steering in mmW hotspot environments, where the user device orientation is modeled as a stochastic variable. Their formulation uses a flat-top antenna model characterized by a pattern function $g(\theta, \phi)$ over the spherical elevation and azimuthal angle coordinates, and minimizes the number of APs required to satisfy a dynamic network coverage constraint. To handle the computational complexity of the joint stochastic placement and beam steering problem, they introduce a greedy algorithm based on the size constrained weighted set cover framework. Their results demonstrate that when the AP beamwidth exceeds $\frac{2\pi}{3}$, the greedy solution matches the optimal AP count, while narrower beams require at most one additional AP.

However, several limitations of this approach motivate our work. Their framework operates at 60 GHz mmW frequencies and does not model THz-specific channel effects such as molecular absorption. The beam steering angles are selected from a small discrete set of spherical elevation and azimuthal values, without jointly optimizing the AP's vertical mounting height. Furthermore, the greedy algorithm provides only an approximate solution with a bounded gap, rather than a provably optimal deployment. Our work extends the concept of azimuth aware deployment to the THz band by including the AP's azimuth orientation ϕ_a as a decision variable within a MILP formulation (Eq. 9), enabling globally optimal beam alignment jointly with 3D spatial coordinate optimization.

E. Research Gap and Our Contribution

Most existing setups rely on flat 2D environments, assume obstacles never move, or settle for basic path-loss equations. Because of this, they struggle to capture the reality of Terahertz (THz) channels. There was no single mathematical formulation that managed to all the necessary real world variables at once like molecular absorption, the nature of dyanmic human blockage and precise directional aiming in a 3D space.

This work steps in to address that exact gap. We have built a unified MILP model designed to optimize both the 3D coordinates and the specific azimuth orientations for THz APs.

Methodology Progression: The overall progression of our methodology is summarized in Fig. 1. Initially, we utilized a graph heuristic-based approach, which was fast but often yielded suboptimal coverage. To guarantee mathematically optimal AP placements, we finally used a robust MILP model. We integrated dynamic human presence probabilities to ensure the network adapts to crowd blockages. Finally, we jointly optimized the azimuth angle orientations to maximize the capabilities of highly directional THz beams as THz beams are highly directional.

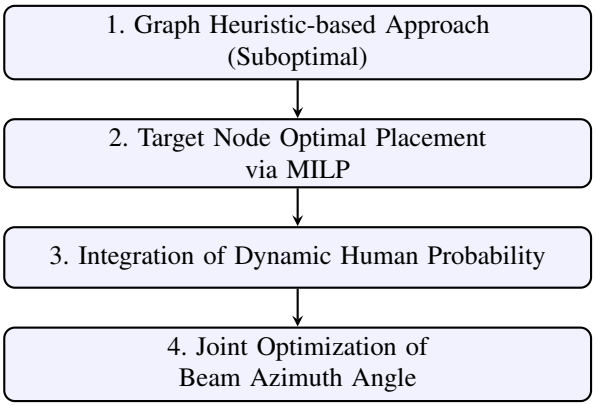


Fig. 1: Progression of the proposed methodology.

III. System Model and Problem Formulation

Designing indoor Terahertz (THz) networks requires strategic placement of Access Points to ensure reliable coverage. The positioning must account for the building's permanent structural layout as well as the random movement of people within it. The key notations used throughout this paper are given in Table I.

TABLE I: Glossary of Proposed Model

Notation	Definition
$\mathcal{U}, \mathcal{A}$	Set of target user locations and candidate AP locations
$d_{u,a}$	3D Euclidean distance between u and a (meters)
$d_{u,a}^{2D}, d_{u,c}^{2D}$	2D projected distances on the horizontal plane (meters)
ϕ_a	Azimuth pointing direction (orientation) of AP a
$\Delta\phi$	Angular deviation between user location and AP orientation
θ_{HPBW}	Half-Power Beamwidth of the directional THz antenna
h_{human}	Average physical height of a human blocker
$z(c)$	Interpolated elevation of the signal beam at grid cell c
$W_a \in \{0,1\}$	1 if an AP is deployed at location a, 0 otherwise
$Y_{u,a} \in \{0,1\}$	1 if user u is associated with AP a, 0 otherwise
η	Target coverage threshold parameter (ratio, e.g., 0.90)
f	Operating frequency (Hz)
c	Speed of light (m/s)
$k(f)$	Molecular absorption coefficient at frequency f
$P_h(u)$	Empirical probability of human presence at location u
$P_{\text{tx}}, P_{\text{rx}}$	Transmit and received signal power (dBm)
$G_{\text{tx}}, G_{\text{rx}}$	Transmit and receive antenna gains (dBi)
$G_{\text{max}}, G_{\text{min}}$	Main-lobe peak gain and side-lobe leakage floor (dBi)
SINR_{th}	Minimum required SINR in each target sub-region
N_0	System noise floor (linear scale)
L_w	Penetration penalty for static walls (dB)
L_h	Penetration penalty for human blockages (dB)
Inf	An arbitrarily large positive constant

The indoor environment is discretized into a 3D volumetric grid. Let $\mathcal{U}$ denote the set of target user equipment locations requiring coverage, and let $\mathcal{A}$ denote the set of candidate AP locations. Unlike traditional 2D placement models, we explicitly optimize over the vertical Z-axis to accurately model signal propagation. Let the 3D coordinates of the user device u be (x_u, y_u, z_u) and the candidate AP a be (x_a, y_a, z_a). The volumetric Euclidean distance $d_{u,a}$ is calculated as:

$$d_{u,a} = \sqrt{(x_u - x_a)^2 + (y_u - y_a)^2 + (z_u - z_a)^2} \quad (1)$$

where the user height z_u is fixed to a standard level, and the optimization algorithm selects an optimal AP mounting elevation z_a from a discrete set of candidate heights.

This 3D distance $d_{u,a}$ denotes the fundamental free-space propagation. The base Free Space Loss (FSL) for a signal operating at frequency f over distance $d_{u,a}$ is derived from the Friis transmission equation:

$$PL_{\text{FSL}}(d_{u,a}) = 20\log_{10}\left(\frac{4\pi f d_{u,a}}{c}\right) \quad (2)$$

where c is the speed of light in a vacuum. The total path loss $PL(u,a)$ for a THz link further incorporates molecular absorption loss and attenuation from physical blockages [1]:

$$PL(u,a) = PL_{\text{FSL}}(d_{u,a}) + PL_{\text{ABS}}(d_{u,a}) + L_{\text{static}} + L_{\text{dynamic}} \quad (3)$$

where $PL_{\text{FSL}}(d_{u,a})$ represents the free-space path loss, L_{static} and L_{dynamic} denote the attenuation penalties from static infrastructure and dynamic human blockages respectively, and $PL_{\text{ABS}}(d_{u,a})$ accounts for atmospheric molecular absorption at high frequencies [2], calculated in dB as:

$$PL_{\text{ABS}}(d_{u,a}) = 4.343 \cdot k(f) \cdot d_{u,a} \quad (4)$$

with $k(f)$ representing the frequency-dependent molecular absorption coefficient.

Due to the highly directional nature of THz signals, line-of-sight (LoS) obstructions severely degrade link quality. We model two distinct types of blockages: static infrastructure and dynamic human mobility. To determine which grid cells and walls lie along the signal path between an AP at (x_a, y_a) and a user at (x_u, y_u), we employ Bresenham's line algorithm. This algorithm efficiently enumerates the set of grid cells $\mathcal{C}_{u,a}$ traversed by the 2D projection of the link by computing the discrete steps along the dominant axis. Specifically, given the horizontal and vertical displacements $\Delta x = |x_u - x_a|$ and $\Delta y = |y_u - y_a|$, Bresenham's algorithm initializes a decision parameter:

$$D_0 = 2\Delta y - \Delta x \quad (5)$$

At each step k along the dominant axis, the next grid cell is selected based on the sign of the decision parameter D_k. If $D_k < 0$, the algorithm advances only along the dominant axis and updates $D_{k+1} = D_k + 2\Delta y$; otherwise, it also steps along the minor axis and updates $D_{k+1} = D_k + 2(\Delta y - \Delta x)$. This rasterization produces the ordered set of grid cells $\mathcal{C}_{u,a}$ through which the signal path passes, without requiring computationally expensive floating-point line intersection calculations.

For static obstacles, let $\mathcal{W}$ denote the set of walls within the environment. If the 3D line segment connecting u and a intersects a wall $w \in \mathcal{W}$ (detected via the grid cells in $\mathcal{C}_{u,a}$), a

material-specific penetration penalty L_w (e.g., concrete, wood, glass) is accumulated:

$$L_{\text{static}} = \sum_{w \in \mathcal{W}_{\text{intersect}}} L_w \tag{6}$$

Furthermore, THz signals suffer severe attenuation when occluded by human bodies [3]. When empirical human footfall datasets are available, we define an expected dynamic loss based on the probability matrix $P_h(c) \in [0, 1]$ for the set of grid cells $\mathcal{C}_{u,a}$ traversed by Bresenham's algorithm along the 2D projection of the link. A human only obstructs the signal if the elevation of the beam $z(c)$ as it passes through cell c drops below the standard human height h_{human}. This geometrical threshold is equivalent to the critical blockage distance limits derived from similar triangles in prior work on THz blockages for humans. Using linear interpolation based on 2D traversed distance $d^{2D}_{u,c}$ and total 2D distance $d^{2D}_{u,a}$, the beam elevation at cell c is given as:

$$z(c) = z_u + \frac{d^{2D}_{u,c}}{d^{2D}_{u,a}}(z_a - z_u) \tag{7}$$

The expected dynamic blockage loss is then defined using an indicator function $\mathbb{I}(\cdot)$, which equals 1 if the beam strikes the human body and 0 otherwise:

$$L_{\text{dynamic}} = \sum_{c \in \mathcal{C}_{u,a}} P_h(c) \cdot L_h \cdot \mathbb{I}\Big(z(c) \leq h_{\text{human}}\Big) \tag{8}$$

where L_h is the empirical penalty caused by human blockage.

To overcome extreme path loss, THz networks use highly directional antennas rather than omnidirectional antennas. We adopt a flat-top sector antenna model parameterized by the half-power beamwidth (HPBW), θ_{HPBW}, and the AP's mounting azimuth orientation ϕ_a. Although our model specifically analyzes the horizontal beamwidth, we assume that APs above the users ($z_a > z_u$) are optimally angled downward. This downward tilt ensures the main signal beam reaches the users directly. As a result, we can simplify our analysis by calculating the signal gain only in the horizontal plane.

Let $\phi_{u,a}$ denote the spatial angle of user u relative to AP a. The shortest angular deviation $\Delta\phi$ between the user's location and the AP's orientation, is calculated as:

$$\Delta\phi = \left|(\phi_{u,a} - \phi_a + 180^\circ) \bmod 360^\circ - 180^\circ\right| \tag{9}$$

This angular deviation ($\Delta\phi$) determines the transmit gain $G_{\text{tx}}(\Delta\phi)$ applied to the link:

$$G_{\text{tx}}(\Delta\phi) = \begin{cases} G_{\text{max}}, & \text{if } \Delta\phi \leq \frac{\theta_{\text{HPBW}}}{2} \\ G_{\text{min}}, & \text{otherwise} \end{cases} \tag{10}$$

where G_{max} is the gain in the main signal direction, and G_{min} represents the small amount of signal leakage in other directions. Since the total radiated energy remains constant, reducing the beamwidth (θ_{HPBW}) increases the main-lobe gain G_{max}. Using APs with narrower beamwidths increases the signal strength and allows communication over longer distances, but the coverage area becomes smaller. G_{min} is modeled as practically negligible, meaning the signal drops significantly outside the HPBW. Finally, the received power $P^{(u,a)}_{\text{rx}}$ (in linear scale) is obtained from the total path loss and the antenna gain as:

$$P^{(u,a)}_{\text{rx}} = 10^{\frac{P_{\text{tx}} + G_{\text{tx}}(\Delta\phi) + G_{\text{rx}} - PL(u,a) - 30}{10}} \tag{11}$$

IV. Proposed Work

Our objective is to increase the minimum required SINR (SINR_{th}) in each target sub-region using the minimum number of THz APs possible.

To achieve this, we formulate our optimization problem as a Mixed-Integer Linear Programming (MILP) model. The primary objective is to minimize the total number of APs deployed, expressed by Eq. (12):

$$\min \sum_{a \in \mathcal{A}} W_a \tag{12}$$

Eq. (13) ensures that each target sub-region is associated with exactly one deployed AP. Eq. (14) mandates that a user u can only logically associate with a candidate location a if an AP is physically deployed there ($W_a = 1$).

$$\sum_{a \in \mathcal{A}} Y_{u,a} = 1, \quad \forall u \in \mathcal{U} \tag{13}$$

$$Y_{u,a} \leq W_a, \quad \forall u \in \mathcal{U}, \forall a \in \mathcal{A} \tag{14}$$

In practical deployment scenarios where satisfying 100% coverage is physically or economically infeasible, Eq. (13) is relaxed to an inequality ($\sum_{a \in \mathcal{A}} Y_{u,a} \leq 1$), and a global coverage constraint is introduced. Let $\eta \in (0, 1]$ denote the required proportion of the total environment that must be served. To strictly enforce this spatial coverage ratio, the total number of successfully associated sub-regions must satisfy the following constraint:

$$\sum_{u \in \mathcal{U}} \sum_{a \in \mathcal{A}} Y_{u,a} \geq \eta |\mathcal{U}| \tag{15}$$

where $|\mathcal{U}|$ represents the total number of target user sub-regions in the partitioned grid.

The next constraint is to maintain the minimum SINR *i.e.*, threshold in each sub-region considering multi AP interference. Eq. (16) ensures that the minimum SINR is maintained in each sub-region:

$$\frac{P^{(u,a)}_{\text{rx}} + \text{Inf} \cdot (1 - Y_{u,a})}{N_0 + \sum_{\substack{b \in \mathcal{A} \\ b \neq a}} P^{(u,b)}_{\text{rx}} W_b} \geq \text{SINR}_{\text{th}}, \quad \forall u \in \mathcal{U}, a \in \mathcal{A} \tag{16}$$

The Left Hand Side (LHS) of Eq. (16) is the ratio of received signal power from the AP to which the user is associated, to the sum of system noise N_0 and interference from other deployed APs. When the uth user is associated with an AP located at a, $Y_{u,a} = 1$. So, the term $\mathit{Inf} \cdot (1 - Y_{u,a})$ goes to zero, and the LHS of Eq. (16) has to satisfy the constraint. Otherwise, if $Y_{u,a} = 0$, the term $\mathit{Inf} \cdot (1 - Y_{u,a})$ takes on an arbitrarily large value. Because of this term, the numerator becomes extremely large and inherently satisfies the

inequality, effectively ignoring the constraint for unassociated links [4]. The term $Inf \cdot (1 - Y_{u,a})$ is needed because a single AP might not provide the minimum SINR to all sub-regions in the building simultaneously; this formulation ensures the overall feasibility of the MILP equations. This MILP formulation is subsequently solved using standard optimization solvers to yield the exact 3D coordinates for optimal AP deployment. The complete source code for both algorithms is publicly available.[1]

Algorithm for AP Placement Without Human Blockage: The first algorithm addresses scenarios where dynamic human mobility is absent. It models terrain hazards from sensor data to identify target and candidate locations, applies an air-to-ground channel model to compute path gains, and then uses a MILP formulation to find the energy-optimal UAV swarm deployment that guarantees signal quality at all targets.

Algorithm for 3D Directional THz AP Placement with Human Blockage: The second algorithm extends the placement to account for dynamic human blockage using the ETH pedestrian dataset. It processes empirical trajectory data to build a human presence probability grid, applies a 3D ray-tracing engine with Bresenham-based blockage detection and directional beamwidth filtering, and uses a MILP formulation to minimize the number of APs while enforcing SINR constraints.

Algorithm 1 Optimal Placement Without Human Blockage

Phase 1: Terrain and Hazard Initialization
Initialize elevation matrix $\mathbf{E}$ and hazard matrix $\mathbf{H}$ of size $N \times N$ to zeros
for each sensor reading $(x, y, v) \in \mathcal{S}$ **do**
  Normalize (x, y) to grid indices (i, j)
  $\mathbf{H}[i][j] \leftarrow \max(\mathbf{H}[i][j], v)$
end for
$\mathbf{H} \leftarrow \mathbf{H} / \max(\mathbf{H})$ {Normalize to $[0, 1]$}
Define targets $\mathcal{T} = \{(i, j, 0) \mid \mathbf{H}[i][j] > 0.2\}$
Define candidates $\mathcal{V} = \{(i, j, z) \mid \mathbf{H}[i][j] > 0,\ z \in [h_{\min}, h_{\max}]\}$

Phase 2: Air-to-Ground Channel Modeling
for each target $t \in \mathcal{T}$ **do**
  for each candidate $v \in \mathcal{V}$ **do**
    $d_{3D} \leftarrow \sqrt{(t_x - v_x)^2 + (t_y - v_y)^2 + (t_z - v_z)^2}$
    $\theta_{elev} \leftarrow \arcsin((v_z - t_z)/d_{3D})$
    $P_{LoS} \leftarrow 1/(1 + a \exp(-b(\theta_{elev} - a)))$
    $L_{FSPL} \leftarrow 20 \log_{10}(d_{3D}) + 20 \log_{10}(f_c) + 32.45$
    $L_{excess} \leftarrow P_{LoS} \times \eta_{LoS} + (1 - P_{LoS}) \times \eta_{NLoS}$
    $G[t][v] \leftarrow 10^{-(L_{FSPL} + L_{excess})/10}$
  end for
end for

Phase 3: Energy-Aware MILP Optimization
Define $X_v \in \{0, 1\}$, $A_{t,v} \in \{0, 1\}$, $P_v \geq 0$
Minimize: $\sum_v (E_{hover} X_v + \alpha P_v)$
subject to:
  $\sum_v X_v \leq K_{\max}$
for each $t \in \mathcal{T}$ **do**
  if $\mathbf{H}[t_x][t_y] \geq 0.8$ **then**
    $\sum_v A_{t,v} \geq 2$ {Redundant monitoring}
  else
    $\sum_v A_{t,v} \geq 1$
  end if
  for each $v \in \mathcal{V}$ **do**
    $A_{t,v} \leq X_v$; $S_{t,v} = G[t][v] \times P_v$
    $M(1 - A_{t,v}) + S_{t,v} \geq \tau_{req}$ {Big-M QoS}
  end for
end for
Solve MILP $\rightarrow \mathcal{V}_{opt} = \{v \mid X_v = 1\}$

Phase 4: Swarm Evaluation
for each $t \in \mathcal{T}$ **do**
  $\mathcal{Q}[t_x][t_y] \leftarrow \max_{v \in \mathcal{V}_{opt}} (G[t][v] \times P_v)$
end for
return $\mathcal{V}_{opt}, \mathcal{Q}$

V. Experimental Setup and Simulation Results

For the experimental setup, we considered two scenarios: first, an indoor academic environment, using the architectural layout of the 4th floor of the Engineering Department 1 at the Indian Institute of Technology (IIT) Bhilai. Second, to capture the stochastic nature of dynamic human mobility, we have used the established empirical ETH Walking Pedestrians dataset in the ETH Zurich academic area to generate realistic spatial blockage probabilities. The environment, network, and optimization parameters used in the simulations are summarized in Table II.

A. Coverage Optimization Under Static Conditions

We first evaluate the formulation in a static environment where dynamic human mobility is absent ($P_h(c) = 0, \forall c$).

Fig. 2 and Fig. 3 illustrate the optimal coordinates selected by the MILP solver to achieve 100% coverage at a SINR threshold of 5 dB. By evaluating candidate AP elevations in discrete 0.2 m increments, the solver achieves this deployment using 14 APs. The 3D volumetric model highlights the solver's ability to optimize over the vertical axis, which helps it adjust the height of the APs to transmit signals through the 1.0 m cubicles and can also be very useful in cases where the ceiling is much higher. Fig. 4 displays the resulting spatial SINR distribution, confirming that this placement strategy successfully covers all areas with the minimum required SINR threshold.

[1] https://github.com/arpitkumar0007/Optimal-AP-Placement-for-THz/tree/main

Algorithm 2 3D Directional THz AP Placement with Human Blockage

Phase 1: Empirical Data Processing
Initialize presence grid $\mathbf{P}$ of size $N \times N$ to zeros
for each trajectory point $(x, y) \in \mathcal{D}$ **do**
  Normalize (x, y) to grid indices (i, j)
  $\mathbf{P}[i][j] \leftarrow \mathbf{P}[i][j] + 1$
end for
$\mathbf{P} \leftarrow \mathbf{P} / \max(\mathbf{P})$ {Normalize to $[0, 1]$}
Define users $\mathcal{U} = \{(i, j, 1.0) \mid \mathbf{P}[i][j] > 0\}$
Define candidates $\mathcal{C} = \{(i, j, z, \theta) \mid \mathbf{P}[i][j] > 0,\ z \in \mathcal{Z},\ \theta \in \Theta\}$

Phase 2: 3D Ray-Tracing & Physics Engine
for each user $u \in \mathcal{U}$ **do**
  for each candidate $c \in \mathcal{C}$ **do**
    $\Delta_{angle} \leftarrow$ angle between c's heading and u
    if $\Delta_{angle} > \phi/2$ **then**
      $P_{RX}[u][c] \leftarrow 0$; **continue**
    end if
    $d_{3D} \leftarrow \sqrt{(u_x - c_x)^2 + (u_y - c_y)^2 + (u_z - c_z)^2}$
    $L_{spread} \leftarrow 20 \log_{10}(d_{3D}) + 20 \log_{10}(f) + 32.45$
    $L_{abs} \leftarrow 4.343 \times k_f \times d_{3D}$
    $\mathcal{L}_{ray} \leftarrow$ BresenhamLine(u_x, u_y, c_x, c_y)
    $L_{block} \leftarrow 0$
    for each cell (x, y) in $\mathcal{L}_{ray}$ **do**
      $z_{ray} \leftarrow$ interpolate ray height at (x, y)
      if $z_{ray} \leq 1.6$ **then**
        L_{block} += $\mathbf{P}[x][y] \times L_h$
      end if
    end for
    $P_{RX}[u][c] \leftarrow 10^{(P_{tx} - L_{spread} - L_{abs} - L_{block})/10}$
  end for
end for

Phase 3: MILP Optimization Engine
Define $W_c, C_u, Y_{u,c} \in \{0, 1\}$
Minimize: $\sum_c W_c$
subject to:
for each $u \in \mathcal{U}$ **do**
  if $\mathbf{P}[u_x][u_y] \geq T$ **then**
    $C_u = 1$ {Must cover high-probability zones}
  end if
  $\sum_c Y_{u,c} = C_u$
  for each $c \in \mathcal{C}$ **do**
    $Y_{u,c} \leq W_c$
    $I_{u,c} = \sum_{k \neq c} P_{RX}[u][k] \times W_k$
    $M(1 - Y_{u,c}) + P_{RX}[u][c] \geq \gamma_{th}(N_0 + I_{u,c})$ {Big-M SINR}
  end for
end for
Solve MILP $\rightarrow \mathcal{A}_{opt} = \{c \mid W_c = 1\}$

Phase 4: Evaluation
for each $u \in \mathcal{U}$ **do**
  $\mathcal{S}[u_x][u_y] \leftarrow \max_{c \in \mathcal{A}_{opt}} \text{SINR}(u, c)$
end for
return $\mathcal{A}_{opt}, \mathcal{S}$

TABLE II: Simulation Parameters

Parameter	Value
Environment Dimensions	32 m × 20 m × 3 m
Operating Frequency (f)	300 GHz
Transmit Power (P_{tx})	20 dBm
Main-Lobe Gain (G_{max})	30 dBi
Side-Lobe Leakage (G_{min})	Negligible ($\leq$ Noise Floor)
Evaluated Beamwidths (θ_{HPBW})	$30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ, 120^\circ$
Candidate Z-axis Elevations (z_a)	2.0 m to 3.0 m (in 0.2 m increments)
Azimuth Rotation Increments	45° ($\pi/4$ rad, from North)
Target Coverage Threshold (η)	0.90 (90%), 0.95 (95%), 1.0 (100%)
Minimum Required SINR (SINR_{th})	5 dB (and 10 dB for strict testing)
System Noise Floor (N_0)	-94 dBm (Linear: 3.981×10^{-10} mW)
Solid Concrete Wall Penalty (L_w)	30 dB
Wooden Door/Cubicle Penalty (L_w)	20 dB
Glass Window Penalty (L_w)	15 dB
Static Cubicle/Desk Height	1.0 m
Human Blocker Penalty (L_h)	20 dB (height 1.6 m)

Fig. 5 illustrates the required number of APs as a function of the minimum SINR threshold for varying spatial coverage targets ($\eta \in \{0.90, 0.95, 1.0\}$). The candidate Z-axis elevations are incremented by 0.2 m ranging from 2.0 m to 3.0 m.When the network demands a higher minimum SINR, the total number of Access Points must increase to compensate for path loss from longer distances and blockage from physical barriers. For a standard coverage requirement of 90% ($\eta = 0.90$), elevating the SINR_{th} from 1 dB to 5 dB forces the MILP solver to increase the AP count from 9 to 10. Furthermore, absolute 100% coverage ($\eta = 1.0$) with threshold of 5 dB requires a peak deployment of 14 APs.

Our primary objective in this scenario is to analyze the fundamental trade-offs between the number of deployed APs, target coverage constraints (η), and the minimum required signal quality (SINR_{th}).

Finally, the choice of antenna direction plays a crucial role in the number of Access Points deployed. It is important to note that varying the Half-Power Beamwidth (θ_{HPBW}) alters the antenna's main-lobe gain (G_{max}) to conserve radiated energy, which directly impacts the Effective Isotropic Radiated Power (EIRP) of the link. Fig. 6 demonstrates the effect of this parameter on the total number of deployed APs, evaluated at a fixed coverage of $\eta = 0.90$ and $\text{SINR}_{\text{th}} = 5$ dB. Utilizing highly directional, narrow beams (60° and 75°) yields higher EIRP and extended linear range, but effectively combining together the dispersed angular areas requires 15 APs. However, as the HPBW widens to 90° and 120°, the EIRP proportionally decreases. Yet, the required number of APs drops sharply to 10. This indicates that in an indoor environment, the spatial advantage of a wider beam that can cover an entire partitioned room from a single coordinate outweighs the pure signal-strength benefits of a higher EIRP.

B. Performance Comparison Against VLC Heuristic

We compare the Deployment efficiency of our proposed 3D MILP Model against a well-known geometric placement

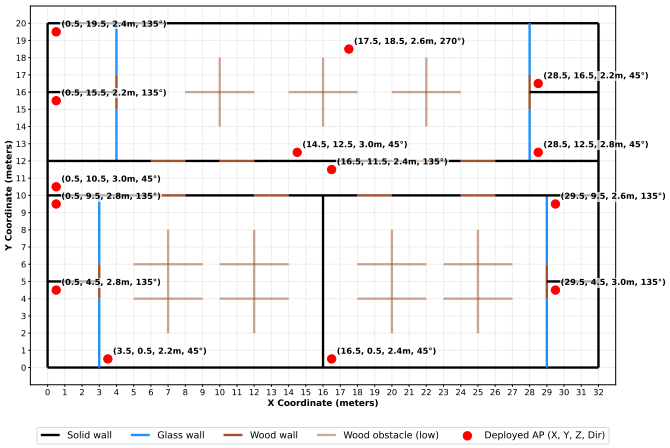


Fig. 2: Optimal 2D projection of spatial coordinates for THz APs in the Engineering Building.

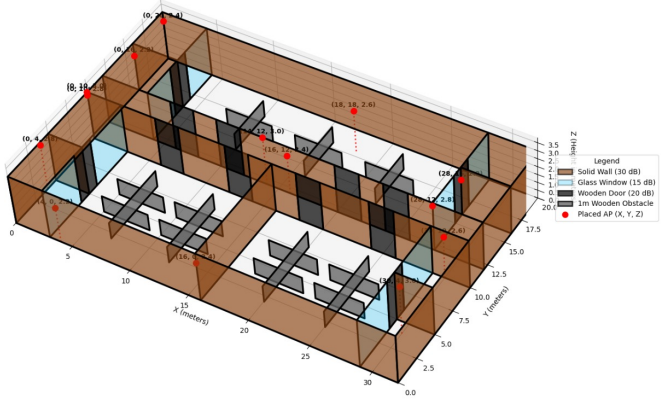


Fig. 3: 3D model of the Engineering Building showing APs.

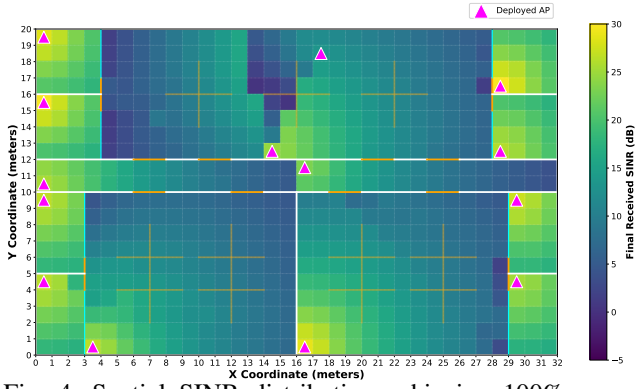


Fig. 4: Spatial SINR distribution achieving 100% coverage (SINR $\geq$ 5 dB).

heuristic. Specifically, we adapted the Iterative Heuristic recently proposed by Shamsi *et al.* [10] for Visible Light Communication (VLC). Because THz and VLC networks share almost the same Line-of-Sight (LoS) propagation constraints, this algorithm can serve as a strong comparison of our model. The heuristic relies on a two-stage process: an Art Gallery Problem, spatial triangulation for coarse coverage, followed by an iterative grid-sampling mechanism to fine-tune the SINR. To ensure equal comparison, both algorithms were given identical environments, physical attenuation rules, and a strict 95% coverage threshold ($\eta = 0.95$).

Fig. 7 shows the total number of APs deployed by both algorithms across varying SINR threshold levels (1 dB, 5 dB, and 10 dB). At lower threshold levels, the MILP model shows a minimal Deployment advantage, needing 12 APs compared to the heuristic's 14 APs at 5 dB. However, under the 10 dB threshold, the heuristic experiences a major reduction in efficiency. Because the heuristic lacks a globally optimal view of the environment, placing subsequent APs causes severe inter-grid interference. To ensure that the SINR threshold is 10 dB, the heuristic approach deploys an excessive 33 APs. Also, the proposed MILP model guarantees a globally optimal layout, concisely aligning the directional beams and using the Z-axis elevation to bypass interference zones. Consequently, the MILP successfully solves requirement utilizing only 17 APs, yielding an approximate 48% reduction in required Deployment compared to the heuristic algorithm.

C. AP Deployment Under Dynamic Human Blockage

Now, we evaluate our 3D deployment strategy in a realistic environment where moving humans block the THz signals. In real buildings, people do not spread out evenly; they tend to gather in groups at specific hotspots or busy corridors. To make a mapping out of these human movements, we use the well-known ETH Walking Pedestrians dataset [9].

Although we previously defined the loss in terms of probability in Eq. (8), here we use a normalized density metric to map dynamic signal blockage across the environment. For the purposes of simulating this dataset, this density metric acts similarly to the blockage probability defined above.

Instead of trying to predict the exact number of people in the building we can use a relative crowd model. This method is highly effective for THz networks because generally the flow of humans remains constant even though the crowd size changes. By focusing on these relative movement patterns, our model can easily identify the crowded areas where Line of Sight (LoS) signals are most likely to be blocked.

To calculate this density metric, we counted the total number of human footfalls denoted as $N(c)$, for each grid cell $c \in \mathcal{C}$. Then, we find the highest-footfall cell in the entire grid and divide each cell by this value to obtain the relative scale. Mathematically, the normalized Human density score for any cell c is:

$$D_h(c) = \frac{N(c)}{\max_{i \in \mathcal{C}} N(i)} \tag{17}$$

This calculation gives every cell a normalized score between 0 and 1. A score of 1 indicates the busiest hotspot, while a score closer to 0 indicates a less crowded area.

Using this score, we introduce a deployment strategy that accounts for dynamic human blockages, controlled by a Density Threshold (τ). If a specific area's density score is equal to or exceeds this threshold ($D_h(c) \geq \tau$), then it is labeled as a "critical" zone. Our formulation then forces the MILP solver to guarantee a signal quality of at least $\text{SINR}_{\text{th}} = 5$ dB specifically in these places where there is more probability of human presence.

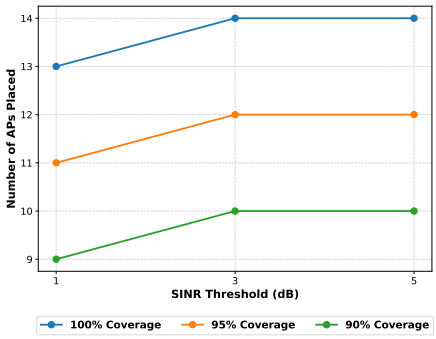


Fig. 5: Required Access Points vs. SINR Threshold for varying coverage.

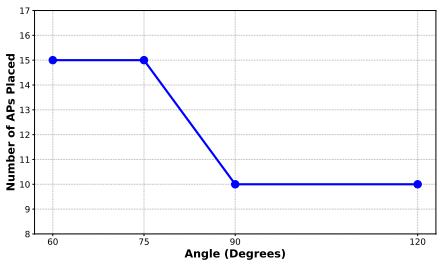


Fig. 6: Number of APs placed vs. Antenna Half-Power Beamwidth for 90% coverage and 5 dB SINR threshold.

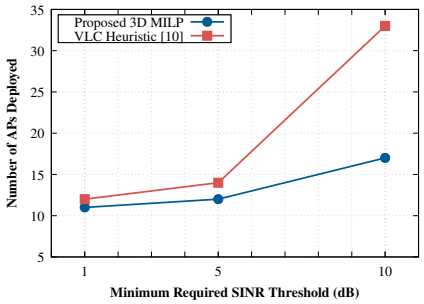


Fig. 7: Deployed APs for the proposed 3D MILP and the VLC Heuristic at a 95% target coverage threshold.

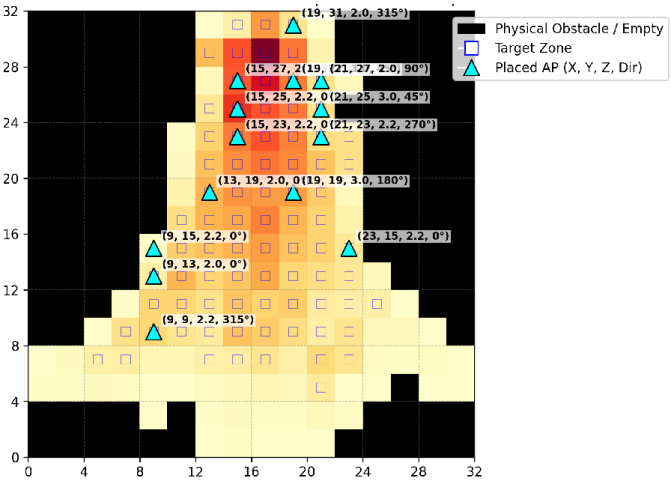


Fig. 8: Optimal AP placement for Density threshold $(\tau) = 0.1$.

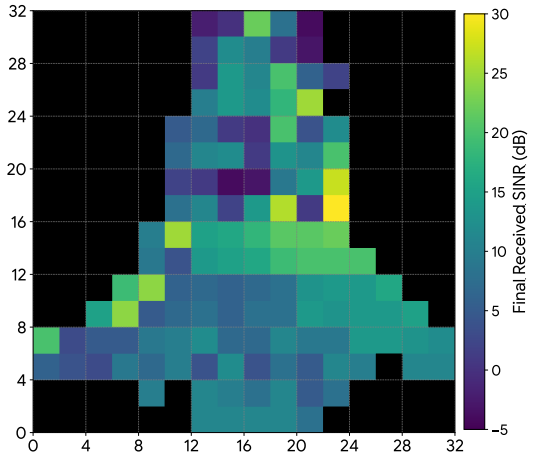


Fig. 9: SINR distribution for density threshold $(\tau) = 0.1$.

Fig. 9 shows us the resulting SINR distribution for this targeted deployment. The heatmap shows that the dynamic blockage-aware metric of our solver guarantees SINR ≥ 5 dB in the critical hotspots/popular spots. In addition, it intentionally allows the signal to drop below the threshold in the less crowded zones. The primary purpose of Fig. 10 is to show how many APs are needed, and what percentage of the floor plan is covered, as we adjust the density threshold (τ). It also serves as a practical decision-making guide to help balance budgets with the required coverage for network operators.

When τ is set to the extremely low value (0.001), the system attempts to cover everything. It requires 26 APs to serve 100.0% of the floor space where humans can go. However, as the threshold increases, the MILP solver stops focusing on empty spaces and allocates APs on the congested hotspots. This results in a significant reduction in the number of required APs. As shown in our base case, raising the threshold to $\tau = 0.100$ decreases the number of required AP counts to 15 while still covering around 60.0% of the area. If we restrict coverage to the absolute highest-density dynamic blockage areas ($\tau = 0.200$), the system requires only 3 APs to cover 41.3% of the area. This demonstrates the scalability of the proposed MILP formulation, enabling Access Point operators to strategically balance the number of deployed APs with coverage in environments with dynamic human blockages.

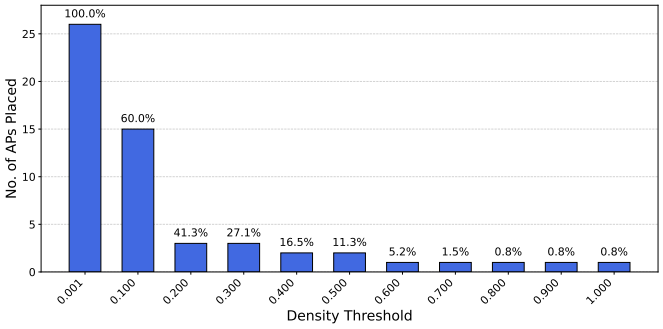


Fig. 10: Number of deployed APs and Total Area Coverage vs. Density Threshold.

D. Impact of Variable AP Heights on SINR

To further validate the importance of volumetric 3D optimization, we compare three distinct height configurations at a strict 100% coverage target with a 5 dB SINR threshold. Fig. 11 presents the cumulative distribution function (CDF)

of the achieved SINR across all grid cells. The coordinates used for this simulation were obtained from the MILP solver's 100% coverage solution. The results demonstrate that uniform AP heights are significantly inferior to variable heights. The 6 m elevation configuration completely fails to deliver the threshold coverage to all cells, leaving 72% of the grid below the minimum required signal level. While the 3 m uniform setup covers all cells, its peak signals remain below 25 dB. In contrast, the variable-height deployment provides peak signals of 30 dB, yielding a 20% increase in maximum signal strength.

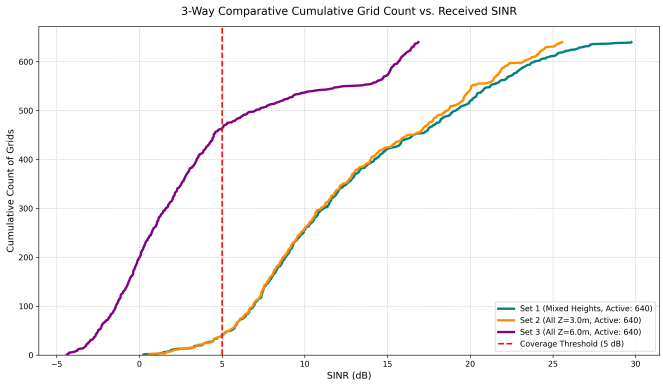


Fig. 11: CDF comparison of SINR distribution for variable height vs. uniform 3 m and 6 m AP deployments at 100% coverage and 5 dB threshold.

VI. Conclusion

To unlock the massive speeds of future 6G Terahertz networks, we must address their biggest physical weakness: signal blockage caused by moving Humans. Current indoor network designs fail because they rely on rigid, brute force AP deployments that ignore the complex realities of the environments they serve. By introducing a rigorous Mixed Integer Linear Programming (MILP) framework, we solved the deployment problem by optimizing the three critical factors: the elevation of the Access Points, the precise directional angles of their beams, and the real-world probability of human footfall. Our mathematical model actively routes signals around static architecture and safely above dynamic human blockages. When put to the test, this optimization strategy guaranteed a strict SINR threshold while getting a massive 48% reduction in required APs compared to the state of the art heuristics. Ultimately, this research shows that the ultra-dense networks of tomorrow will not be built by simply installing more equipment, but by mathematically weaving signals through the living crowds they connect.

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